Clinical Imaging & Verification Report

Uncertainty-Calibrated & Adaptive Self-Verifying Pulmonary Assessment

REPORT
2026-10-19

PATIENT & SCAN PROFILE

PATIENT ID	SCAN DATE	SCAN FORMAT	SLICES
Lung_Dx-B0041	2026-10-19	Real DICOM Slice (C:\Users\spide\Desktop\LUCID-CT\1.dcm)	16

ESTIMATED CATEGORY

4A Suspicious

Calibrated Malignancy Risk: 44.9%

RISK TIER
INTERMEDIATE

Estimated diameter: 10.0 mm

ADAPTIVE VERIFICATION & RELIABILITY GATE

ADAPTIVELY_RESOLVED | {Benign} (90%)

ADAPTIVELY_RESOLVED

Recommended Clinical Action

VERIFIED: Single-label {Benign} guaranteed at 90.0% statistical coverage (Mondrian stratum: MEDIUM_QUALITY, nonconformity: 0.577 <= threshold 0.600). Benign appearance confirmed; annual LDCT screening recommended.

TARGETED RE-ANALYSIS AUDIT TRAIL

Agreement with Baseline: 99.2%

* 1.25x High-Magnification Boundary Shell Zoom

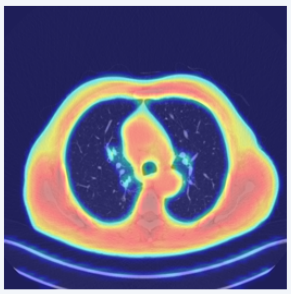
UNCERTAINTY & VERIFICATION INDICES

Epistemic Uncertainty	**0.0109**
Aleatoric Uncertainty	**0.6650**
Predictive Entropy	**0.6874**
Scan Quality Score	**73.9 / 100**
Grad-CAM Fidelity	**0.65 / 1.0**
Trait Physical Grounding	**0.67 / 1.0**

RADIOLOGICAL TRAIT EVIDENCE

Spiculation	**2.29**
Lobulation	**2.26**
Calcification	**3.46**
Subtlety	**3.57**
Margin	**2.06**
Sphericity	**2.27**

XAI LOCALIZATION & EVIDENCE GROUNDING



Estimated from diameter alone, assuming a baseline solid nodule. Nodule type, growth, and prior imaging were not available.